

Effects of EPX, IMD and BOS on *A. caliginosa* - Preliminary tests

Protocol

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Objective : To identify the concentration ranges for each molecule to test that will enable the construction of dose-response curves for growth and reproduction.

1 Generalities

1.1 Earthworms

Earthworms for these preliminary tests are *A. icterica* individuals fetch in a meadow in Versailles (Les Closeaux). *A. caliginosa* were unavailable and *A. icterica*, a similar species, is much easier to fetch in Versailles. The use of this species will still allow for the identification of concentration ranges to test for the real EC₅₀ tests.

1.2 Design

Table 1.1: Primary solutions concentrations for each molecule in acetone

molecule	concentration_p1	concentration_p2
EPX	3000	600
IMD	300	60
BOS	3000	600

Cosms are made with 200 g of Closeaux soil and earthworms are fed with 3 g/ind/14d.

Each modality is made with 1 replicates for growth and 1 replicates for reproduction.

The growth test lasts 28 days and the reproduction test lasts 28 days.

Table 1.2: Tested concentrations for each molecule (mg/kg)

EPX	IMD	BOS
1e-02	1e-03	1e-02
1e-01	1e-02	1e-01
1e+00	1e-01	1e+00
1e+01	1e+00	1e+01
1e+02	1e+01	1e+02

15 concentrations tested.

For the growth test, one replicate correspond to 2 individuals per cosm. For the reproduction test, one replicate correspond to 2 individuals per cosm.

1.3 Number of individuals and number of cosms

Table 1.3: Number of needed individuals

Experience	EPX	IMD	BOS	Total
Growth	10	10	10	30
Reproduction	10	10	10	30
Total	20	20	20	60

Table 1.4: Number of needed cosms

Experience	Nb_cosm
Growth	15
Reproduction	15
Total	30

1.4 Soil and horse dung needs

Cosms preparation needs 6 kg of Closeaux soil. 360 g of horse dung are needed in total.

2 Protocol

2.1 Calculations

2.1.1 Preparations for EPX

Table 2.1: Soil preparation for each EPX concentration

Concentration (mg/kg)	Q_{EPX} 1conc (mg)	Q_{tot} sol (g)	Veau (CRE 70%)	Vsol.p1 (μ L)	Vsol.p2 (μ L)	Choice Sol.p
0.01	0.004	400	548.8	1.3333	6.67	2
0.10	0.040	400	548.8	13.3333	66.67	2
1.00	0.400	400	548.8	133.3333	666.67	2
10.00	4.000	400	548.8	1333.3333	6666.67	2
100.00	40.000	400	548.8	13333.3333	66666.67	1

13.3333333 mL of primary solution n°1 are needed.

7.4066667 mL of primary solution n°2 are needed..

20 mL of primary solution n°1 are prepared with 60 mg of EPX in Acetone. 10 mL of primary solution n°2 are prepared with 2 mL of the primary solution n°1 in Acetone.

2.1.2 Preparations for IMD

Table 2.2: Soil preparation for each IMD concentration

Concentration (mg/kg)	Q_{IMD} 1conc (mg)	Q_{tot} sol (g)	Veau (CRE 70%)	Vsol.p1 (μ L)	Vsol.p2 (μ L)	Choice Sol.p
0.001	0.0004	400	548.8	1.3333	6.67	2
0.010	0.0040	400	548.8	13.3333	66.67	2
0.100	0.0400	400	548.8	133.3333	666.67	2
1.000	0.4000	400	548.8	1333.3333	6666.67	2
10.000	4.0000	400	548.8	13333.3333	66666.67	1

13.3333333 mL of primary solution n°1 are needed.

7.4066667 mL of primary solution n°2 are needed..

10 mL of primary solution n°2 are prepared with 2 mL of the primary solution n°1 in Acetone.

2.1.3 Preparations for BOS

Table 2.3: Soil preparation for each BOS concentration

Concentration (mg/kg)	Q_{BOS} lconc (mg)	Q_{tot} sol (g)	Veau (CRE 70%)	Vsol.p1 (μ L)	Vsol.p2 (μ L)	Choice Sol.p
0.01	0.004	400	548.8	1.3333	6.67	2
0.10	0.040	400	548.8	13.3333	66.67	2
1.00	0.400	400	548.8	133.3333	666.67	2
10.00	4.000	400	548.8	1333.3333	6666.67	2
100.00	40.000	400	548.8	13333.3333	66666.67	1

13.3333333 mL of primary solution n°1 are needed.

7.4066667 mL of primary solution n°2 are needed..

20 mL of primary solution n°1 are prepared with 60 mg of BOS in Acetone. 10 mL of primary solution n°2 are prepared with 2 mL of the primary solution n°1 in Acetone.

2.2 D-4 - Soil preparation

In the morning, around 6 kg of soil are put in steel recipient in a thermoregulated chamber at 110°C.

In the afternoon, in 15 inox recipient, 400 g of this soil are weighted and 548.8 mL of ISO water are added (do not mix !).

2.3 D-3 - Soil contamination

Cosms left from the first try are taken out from the refrigerated chamber at -20°C and are put in the thermoregulated chamber at 18°C.

Primary solutions are prepared.

For each molecule and each concentrations :

- The volume given by Table 2.3, Table 2.2 or Table 2.1 is put in a 10 mL beaker
- Acetone is added to have a volume of around 10 mL
- This solution is directly put on the soil while mixing with a seringue

- The cosms for the given concentration are prepared with 255 g of this soil preparation
- Cosms are left openn under a hood.

3 D0 - Eathworms introduction

Horse dung is added and mixed to each cosm : 3 g/ind (dry weight) with its WHC adjust to 70%.

Juveniles earthworms are individually weighted and put into their cosms by two. For each reproduction cosm, two adults are weighted together and then put into the cosm.

The weight of the cosms are recorded to be able to adjust the soil humidity over time.

4 Maintenance

Soil humidity is checked each week. Juveniles earthworms are weighted on D14. All earthworms are fed with 3 g/ind (dry weight) with the WHC adjust to 70% on D14.

5 D28 - End of exposition

All earthworms are weighted.

Cocoons from the reproduction cosms are retrieved using a water sieving technique described in Bart et al. (2019).